MILITARY SPECIFICATION

INTERFACE, TELEPHONE SIGNALING TA-978()/G

This specification is approved for use by the Communications Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of 2-wire Telephone Signaling Interface (TSI) unit designated as Interface, Telephone Signaling TA-978()/G. (See 6.1 and 6.6)

2. APPLICABLE DOCUMENTS

2.1 Issues of Documents. The following documents of the issue in effect on the date of invitation for Bids or Request for Proposal, form a part of this specification to the extent specified herein:

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Communications Research and Development Command, ATTN: DRDCO-CM-DM, Fort Monmouth, NJ 07703 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5810

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SPECIFICATIONS

Military

MIL-P-116	Preservation-Packaging, Methods of
MIL-P-11268	Parts, Materials, and Processed Used in Electronic Equipment
MIL-M-13231	Marking of Electronic Equipment
MIL-F-14072	Finishes for Ground Electronic Equipment
MIL-E-55585	Electronic Equipment and Parts (pack)

STANDARDS

Federal

FED-STD-141	Paint, Varnish, Lacquer and Related Materials, Methods of Inspection, sampling and testing

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment Subsystems, and Systems
MIL-STD-462	Electromagnetic Interference Characteristics Measurement of
MIL-STD-726	Packaging Requirements Code
MIL-STD-781	Reliability Tests, Exponential Distribution

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MIL-STD-810	Environmental Test Methods
MIL-STD-1275	Characteristics of 28V DC Electrical System in Military Vehicles

DRAWINGS/SPECIFICATIONS

DL-SM-B-944400	Interface, Telephone Signaling TA-978( )/G
MIL-N-49205	Nuclear Survivability Requirements For Interface, Telephone Signaling TA-978( )/G (Confidential)
SM-D-944491	Cable Assembly, NESTOR
SM-D-930717	Cable, Special Purpose, CX13072/u Assembly
SM-D-944457	PW Assembly Extender Board
SC-A-362100	Requirements for Preparation of Packaging and Packing Illustrations and Bills of Material

PUBLICATIONS

National Security Agency

CSEEB-18	Communications Security Equipment Engineering Bulletin For TSEC/KY-38 (Confidential)
CSESD-14	Communications Security Equipment System Document For TESC/KY-57/58 (Confidential)

3. REQUIREMENTS

3.1 First Article. When specified in the contract or purchase order, the contractor shall furnish First Article Units in accordance with 4.3.

3.2 Construction. The equipment shall be constructed in accordance with the requirements of this specification and Drawing and Data List DL-SM-B-944400, (including the requirements for parts, materials and processed thereon) (See 4.4).

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3.3 Parts, Materials and Processes, General. In addition to the requirements of this specifications, the requirements of MIL-P-11268, including the selection requirements therein, shall apply (See 4.4).

3.4 Finish. The equipment shall be finished in accordance with MIL-F-14072, and the equipment drawings (See 4.4).

3.5 Marking.

3.5.1 General. Marking shall conform to Spec MIL-M-13213. Panel markings shall be group 1 (See 4.4).

3.5.2 Visability. Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of equipment (See 4.4).

3.5.3 Serial Numbers. The equipment shall be serially numbered.

3.6 Interchangeability. Dimension of TA-978()/G shall be as specified in the applicable drawings; and like units, assemblies, subassemblies, and replaceable parts shall meet Requirement 7 of MIL-STD-454 (See 4.9.1).

3.7 Preconditioning. (See 4.6)

3.7.1 Bounce Reconditioning. Each equipment shall meet the requirements specified herein, without subsequent processing, after subjection to the bounce preconditioning of 4.6.1.

3.7.2 Electrical Preconditioning (burn-in). Prior to submission to the specified quality conformance inspections, each unit shall be subjected to a minimum 48 hour on-time burn-in period under the conditions specified for reliability testing. The burn-in period shall be concluded by a failure free period of 24 hours (See 4.6.2).

3.8 Electrical

3.8.1 Audible and Visual Alarms. (See 4.8.11)

3.8.1.1 Audible Alarm. The audible alarm shall be an acoustic tone of 1850 ± 100 Hz at a level of 32 ± 5dBa with the volume control in the fully counterclockwise position and a level of 88dBa or greater acoustically measured at a distance of three feet with the volume control in the fully clockwise position (See 4.8.11.2).

3.8.1.2 Visual Alarms. (See 4.8.11.1)

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3.8.1.2.1 Call Alert (AMBER). The visual alarm shall have a minimum brightness of 4 Ft-Lamberts direct view photoptic response averaged over the entire area (See 4.8.11.1).

3.8.1.2.2 Cypher Text (Green). The visual alarm shall have a minimum brightness of 4 Ft-Lamberts direct view photoptic response averaged over the entire area (See 4.8.11.3).

3.8.2 DTMF Signaling (See 4.8.8)

3.8.2.1 Output Level. (See 4.8.8.1)
The composite dual tone output level shall be -4dBm ± 2dB.

3.8.2.2 Frequency. (See 4.8.8.1)

Key	Frequency	
1	697/1209	±1.3%
2	697/1336	"
3	697/1477	"
4	770/1209	"
5	770/1336	"
6	770/1477	"
7	852/1209	"
8	852/1336	"
9	852/1477	"
0	941/1336	"
FO	697/1633	"
F	770/1633	"
I	852/1633	"
P	941/1633	"
R	941/1209	"
C	941/1477	"

3.8.2.3 Sidetone. The sidetone output level shall be -21 ± 2dBm (See 4.8.8.2).

3.8.3 Plain Text Impedance.

3.8.3.1 Wire Line Interface Impedance. In plain text mode the impedance presented by the TA-978()/G to the wire line, in receiving and transmitting, shall be 600 ohms ±20% at the frequency of 1kHz (See 4.8.3.1).

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3.8.3.2 Secure Voice Unit Interface Impedance. In plain text mode the impedance presented by the TA-978()/G to the plain text output of the secure voice unit shall be 150 ohms ±20% at the frequency of 1 kHz (See 4.8.3.2).

3.8.4 Cipher Text Impedance. (See 4.8.4)

3.8.4.1 Wire Line Interface Impedance. In cipher text mode the impedance presented by the TA-978()/G to the wire line shall be 150 ohms ±20% in receiving and 75 ohms ±20% in transmitting, both at the frequency of 16 kHz (See 4.8.4.1).

3.8.4.2 Secure Voice Interface Impedance. In cipher text mode the impedance presented by the TA-978()/G to the cipher text output of the secure voice unit shall be greater than 2000 ohms at the frequency of 16 kHz (See 4.8.4.2).

3.8.5 Ringer Output. The ringer output signal to the wire line at the frequency of 20 Hz ±20% with the output isolated from equipment ground shall be as specified in Figure 1 (See 4.8.5).

3.8.6 Ring Detector. The TA-978()/G shall detect a 20 Hz ±20% ring signal of 25 v rms or greater in the on-hook position (See 4.8.6).

3.8.7 Common Battery Supervision (CBS). The TA-978()/G in the off-hook position shall present to the wire line a dc loop closure resistance of maximum 450 ohms. The impedance of the CBS element shall be at least 6000 ohms for frequencies equal to or greater than 200 Hz. In the on-hook position, the CBS element shall be disconnected from at least one side of the wire line (See 4.8.7).

3.8.8 Plain Text Signal Characteristics. (See 4.8.9)

3.8.8.1 TA-978()/G to Wire Line. (See 4.8.9.1)

3.8.8.1.1 Signal Level, Frequency Response & Distortion. The TA-978()/G shall present to the wire line a plain text signal of -4dBm±3dB at frequencies of 300 Hz, 1kHz, 2 kHz and 3.5 kHz. The distortion of the plain text signal shall be no more than 5% (See 4.8.9.1.1).

3.8.8.1.2 Sidetone Signal Level. The sidetone level at the handset output shall be 12 ±2db below the wire line signal level at the frequency of 1 kHz (See 4.8.9.1.2).

3.8.8.2 TA-978()/G to Secure Voice Device. (See 4.8.9.2)

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3.8.8.2.1 Signal Level, Frequency Response & Distortion. The TA-978()/G shall have a signal gain of 0 ±1.5db for received signals of -20dBm and +1dBm at frequencies of 300Hz, 1kHz, 2kHz and 3.5kHz. The distortion of the plain text signal shall be no more than 5% (See 4.8.9.2.1).

3.8.9 Cipher Text Signal Characteristics. (See 4.8.10)

3.8.9.1 TA-978()/G to Wire Line. (See 4.8.10.1)

3.8.9.1.1 Signal Level and Risetime. The TA-978()/G shall provide a cipher text signal of 3.0 ±0.5 V peak to peak to the wire line. The rise and fall times of the cipher text signal shall not exceed 10 microseconds between the 10% and 90% voltage levels of the signals (See 4.8.10.1.1).

3.8.9.2 TA-978()/G to Secure Voice Device. (See 4.8.10.2)

3.8.9.2.1 Low Signal Level. The TA-978()/G shall provide a cipher text signal equal to or greater than 0.5V peak to peak to the secure voice device (See 4.8.10.2.1).

3.8.9.2.2 High Signal Level. The TA-978()/G shall provide a cipher text signal equal to or greater than 7.0V peak to peak to the secure voice device (See 4.8.10.2.2).

3.8.9.2.3 Distortion. For signals received from the wire line of 10, 20, 50, 75 and 100 kHz the distortion of the signal provided to the secure voice device cipher text input shall not exceed 5% (See 4.8.10.2.3).

3.8.10 Power Requirements. (See 4.8.12)

3.8.10.1 Battery Voltage. The TA-978()/G shall meet all specified performance requirements when powered by a battery voltage varying from a minimum of 20 volts dc to a maximum of 32 volts dc (See 4.8.12.1).

3.8.10.2 Battery Power. (See 4.8.12.2)

3.8.10.2.1 Maximum Power Average. The power consumption of the TA-978()/G shall not exceed an average of 3.0 watts (excluding power to NESTOR device) with the battery supply between 20 and 32 volts dc (See 4.8.12.2.1).

3.8.10.2.2 Standby Power. With the handset "on-hook" and the power switch to "on", the TA-978()/G shall not consume any power when operated at 28 volts dc. With handset "on-hook" and operated at 32 volts dc, the operating current shall not exceed 15 mA (See 4.8.12.2.2).

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3.8.10.3 Vehicular Power. The TA-978()/G shall meet all specified performance requirements when connected to the vehicular power adapter HYP-57, with the VPA powered from a vehicular power source varying from a minimum of 22 volts dc to a maximum of 32 volts dc (See 4.8.12.3).

3.8.10.4 Transient Voltage Protector. The TA-978()/G shall meet the vehicle equipment power requirements specified in MIL-STD-1275(AT) (See 4.8.12.4).

3.8.10.5 NESTOR Power. The TA-978()/G shall provide the required power for the TSEC/KY-38 Secure Voice Equipment, as specified in CSEEB-18, under all conditions of battery and vehicular power (See 4.8.12.5).

3.8.10.6 Reverse Polarity Protection. The TA-978()/G shall not sustain any damage when connected to a battery or vehicular power source of reversed polarity and shall meet all specified performance requirements after polarity is reversed to normal (See 4.8.12.6).

3.8.10.7 Surge Protection. The TA-978()/G shall have protection devices that protect operating personnel from extraneous high voltage surges (lightning and nuclear EMP) appearing at the input and output terminals of the equipment (See 4.8.13).

3.8.11 Operational Performance Test (See 4.15) The TA-978()/G shall provide satisfactory performance when tested in accordance with para 4.15.

3.9 Enviormental. The equipment shall meet the following enviormental conditions (See 4.7).

3.9.1 Temperature. (See 4.7.1)

3.9.1.1 Low. During and after testing as specified in 4.7.1.1, the equipment shall meet the requirements of 3.8.11.

3.9.1.2 High. During and after testing as specified in 4.7.1.2, the equipment shall meet the requirements of 3.8.11.

3.9.2 Humidity. During and after testing as specified in 4.7.3, the equipment shall exhibit no physical damage, such as corrosion, rust, blistering, swelling, or deterioration of parts and materials, and shall meet the requirements of 3.8.11.

3.9.3 Altitude. The equipment shall show no evidence of damage or deterioration in performance at altitudes up to 15,000 feet above sea level and after storage at altitudes up to 40,000 feet above sea level (See 4.7.2).

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3.9.4 Leakage (Immersion). The equipment shall show no evidence of leakage after testing as specified in 4.7.7.

3.9.5 Shock, Drop. Following testing as specified in 4.7.6, the equipment shall meet the requirements of 3.8.11 and any physical damage shall be minor as specified in MIL-STD-252.

3.9.6 Shock, Bench Handling. Following testing as specified in 4.7.5, the equipment shall meet the requirements of 3.8.11 and shall sustain no physical damage.

3.9.7 Bounce, Loose Cargo. Following testing as specified in 4.7.4.2, the equipment shall meet the requirements of 3.8.11 and shall sustain no physical damage which would impair operation.

3.9.8 Salt Fog. After the salt fog test as specified in 4.7.9, the equipment when examined visually with the aid of 10 power magnifier, shall show no evidence of degradation, such as flaking, pitting, blistering or loosening of finish or metal surface, or exfoliation of metal (See 6.8.6).

3.9.9 Vibration. When tested in accordance with 4.7.4.1, the equipment shall meet the requirements of 3.8.11.

3.9.10 Dust. When tested in accordance with 4.7.8, the TA-978()/G shall meet the requirements of 3.8.11.

3.9.11 Fungus. After the fungus test of 4.7.10, the TA-978()/G shall be visually examined using a 10 power magnifier. The equipment shall show no more than six minute unrelated spots, each no greater than .015 square inches in area, of sparse microbial growth as evidenced by growth colonization which includes branching and sporulation on or within each cubic foot, or fraction thereof, of equipment assembly volume. Isolated instances of partial tabular germination shall not be included in this evaluation (See 6.8).

3.10 Electromagnetic Interference. (See 4.10)

3.10.1 Electromagnetic Compatibility. The TA-978()/G shall comply with the following electromagnetic interference requirements specified in MIL-STD-461. Notice 4, (See 4.10.1).

CE01	CS01	RE02	RS02
CE03	CS02	RE02.1	RS03
CE04	CS06		
CE05			

The upper frequency unit of test for RE02.1 shall be 1GHz.

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3.10.1.1 Production Testing. For subsequent group C inspection only, CS01, RE02, RE02.1 and RS03 EMI tests are required. The upper frequency limits for RE02, RE02.1 and RS03 tests shall be 200 MHz.

3.10.2 Bonding and Grounding. The DC resistance of all bonds and grounds (including connector shell to the case or chassis) shall not exceed 2.5 milliohms (See 4.10.2).

3.11 Reliability. The equipment shall demonstrate a specified mean-time-between-failure (MTBF) of ten-thousand (10,000) hours when tested in accordance with 4.11.

3.12 Nuclear Survivability. (See 4.12). The TA-978( )/G shall sustain no damage, shall continue to operate and shall meet all specified performance requirements of 3.8.11, before and after being exposed to the nuclear environmental conditions of specification MIL-N-49205(CR).

3.13 Air Seal Test. The TA-978( )/G shall be subjected to the Air Seal Test, and shall not exceed 0.01 pound/minute decrease in vacuum/pressure (See 4.17).

3.14 Specular Gloss. The 60-degree specular gloss of the equipment case shall be less than 30 (See 4.20).

3.15 Workmanship. The TA-978( )/G shall be manufactured and assembled in accordance with requirement 9 of MIL-STD-454 and the applicable portions of MIL-P-11268 (See 4.4).

3.16 Electrical Performance of Assemblies (Modules) and Enclosure Assembly. Individual assemblies shall be tested to determine compliance with the performance requirement specified in 4.8.

3.17 Systems Safety Requirements.

3.17.1 Personnel Hazards. Personnel hazards shall be kept to a minimum. The criteria of MIL-STD-454, requirement 1, shall be made applicable for tasks and efforts such as the selection of parts, the complete manufacturing and assembly process, and any product baseline configuration changes that may be implemented during the course of the contract. Verification that compliance with this requirement has been achieved shall be through a visual inspection (See 4.13).

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Test equipment and facilites for Government Inspections shall be made available by the contractor. The Government reserves the right for authorized technical representatives to witness any contractor performed inspections and tests.

4.2 Classification of Inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection. (See 4.3) Does not include preparation for delivery.

b. Inspections covered by subsidiary documents. (See 4.4)

c. Quality Conformance inspections.

(1) Quality conformance inspection of equipment before packaging. (See 4.5)

(2) Quality conformance inspection of packaging. (See 4.14)

4.3 First Article. Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor.

4.3.1 First Article Units. The contractor shall furnish the number of first article units in accordance with the following:

a. Sixteen (16) each TA-978()/G

b. Six (6) each cable SM-D-944491*

c. Ten (10) each cable SM-D-930717*

4.3.2 First Article Inspection. The first article inspection shall consist of inspections specified in Table IV and shall be performed in the order specified therein.

* Cable lengths to be specified in contract or purchase order.

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Table IV. First Article Inspection

Inspections	Reqt Para	Test Para	Unit 1	Unit 2	Unit 3	Unit 4	Unit 5	Unit 6	Units 7 thru 16
1. Inspections Covered by Subsidiary Documents		4.4	Inspections to be performed on all units						
2. Bounce Preconditioning	3.7.1	4.6.1	Inspections to be performed on all units						
3. Electrical Preconditioning	3.7.2	4.6.2	"	" "	"	" "	"		
4. Group A Inspection	(See Table V) 3.14	4.20	"	" "	"	" "	"		
5. Group B Inspection	(See Table VI)		"	" "	"	" "	"		
6. Group C Inspection a),b),c)									
Temperature	3.9.1	4.7.1	1						
Altitude	3.9.3	4.7.2	2						
Humidity	3.9.2	4.7.3	3						
Vibration	3.9.9	4.7.4.1		2					
Bounce, loose cargo	3.9.7	4.7.4.2		3					
Shock, bench handling	3.9.6	4.7.5		4					
Shock, drop	3.9.5	4.7.6		5					
Leakage (immersion)	3.9.4	4.7.7		1&6					
Dust (fine sand) d)	3.9.10	4.7.8			1				
Salt Fog d)	3.9.8	4.7.9			2				
Fungus d)	3.9.11	4.7.10				1			
EMI	3.10	4.10					1		
Nuclear Survivability e)	3.12	4.12			3(back up)			1	
7. Group D reliability a), b), c)	3.11	4.11							1

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TABLE IV

NOTES:

a. The inspections 1 through 5, in the order shown, shall be performed on all first article units, except as noted in Table VI , before subjecting these units to any other inspection requirements specified in the table.

b. The implementation of inspections covered by 6 and 7 shall be expedited to satisfy contract delivery requirements.

c. The numbers in the unit columns in the table covered by 6 and 7 specify the order of inspections for the indicated units.

d. The equipment shall be thoroughly washed, cleaned, dried and refurbished before proceeding with subsequent inspection.

e. If required by contract or purchase order, after completion of Group A and Group B tests, one unit shall be subjected to the nuclear survivability test of paragraph 4.12 to demonstrate compliance with the requirements of paragraph 3.12.

4.4 Inspections Covered by Subsidiary Documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification, and the inspection requirement specified in the contract or purchase order.

Item	Where Required
Construction	3.2
Parts, materials, and processed, general	3.3
Finish	3.4
Marking	3.5
Workmanship	3.15

4.5 Quality Conformance Inspection of Equipment before Preparation for Delivery. The contractor shall perform the inspection specified in 4.4, 4.5.1 through 4.5.4 and 4.6. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

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4.5.1 Group A Inspection. Subassemblied and completed equipments shall be inspected for conformance to the inspections specified in Table V. Each subassembly after 100% inspection, shall be formed into discrete lots for quality assurance verification. The equipments assembled from lots accepted under subassembly verification shall be inspected 100% and formed into discrete lots for final QA verification. Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be submitted to sampling inspection, utilizing the procedures of MIL-STD-105 using general inspection levels and AQL's indicated in Table V (Quality Assurance Verification).

4.5.1.1 Order of Inspection within Group A. Group A inspection shall be performed in an order satisfactory to the Government, except that the operational test in 4.15 shall be next to last and the air-seal test in 4.17 shall be last.

4.5.2 Group B Inspection. Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection. This inspection shall conform to Table VI and to the special inspection levels of Table 1 of MIL-STD-105.

4.5.2.1 Group B Sampling Plans. The Group B AQL shall be as shown in Table VI and the inspection level shall be S-4.

4.5.2.2 Order of Inspection within Group B. Within Group B inspection interchangeability (dimensional) inspection shall be performed first, followed by the electrical tests in an order satisfactory to the Government.

4.5.3 Group C Inspection. Group C inspection shall be performed on units that have passed Group A and Group B inspection. The inspection shall consist of the inspection specified in Table VII. Samples shall be selected in accordance with 4.5.3.1.

4.5.3.1 Sampling for Group C Inspection.

4.5.3.1.1 Subgroup 1. For this subgroup, two units shall be selected at random from the first production lot. For subsequent Group C inspection , two units shall be selected from one month's production or 100 units whichever is smaller. One unit shall be subjected to leakage (immersion), temperature, vibration, and leakage (immersion) testing, and one unit shall be subjected to altitude and humidity testing.

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Table V Group A Inspection

Inspection	Requirement Paragraph	Inspection Paragraph	AQL Major	AQL Minor
Visual-Mechanical	3.15, 3.6, 3.17	4.13		
Printed Wiring Assemblies			0.65%	2.5%
Assembled Top Panel			0.65%	2.5%
Assembled Front Panel			0.65%	2.5%
TA-978()/G Enclosure Assembly less plug-in modules			1.0 %	4.0%
Assembled TA-978()/G			1.0 %	4.0%
Electrical				
Assembly/Sub Assembly				
Printed Wiring Assemblies	3.16	4.18	1.0%	
Assembled Top Panel	3.16	4.18	1.0%	
Assembled Front Panel	3.16	4.18	1.0%	
TA-978()/G Enclosure Assembly less plug-in modules	3.16	4.18	1.0%	
Air-Seal	3.13	4.17	1.0%	
System Operational Test				
Operational Performance	3.8.11	4.15	1.0%	

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Table VI Group B Inspection

Inspection	Reqt Para	Test Para	AQL
Interchangeability (Dimensional (modules)	3.6	4.9.1	4.0%
Electrical			
Impedances	3.8.3, 3.8.4	4.8.3, 4.8.4	
Ringer Output	3.8.5	4.8.5	
Ring Detector	3.8.6	4.8.6	
Common Battery Supervision (NOTE 1)	3.8.7	4.8.7	
DTMF Signaling	3.8.2	4.8.8	
Plain Text Signal Characteristics	3.8.8	4.8.9	
Cipher Text Signal Characteristics	3.8.9	4.8.10	
Alarms (NOTE 2)	3.8.1	4.8.11	
Power Requirements, Battery	3.8.10.1 & 2	4.8.13.1 & 2	
Power Requirements, Vehicular	3.8.10.3 & 4	4.8.13.3 & 4	
Power For Secure Voice Equipment	3.8.10.5	4.8.12.5	
Reverse Polarity Protection	3.8.10.6	4.8.12.6	
Surge Protection (NOTE 3)	3.8.10.7	4.8.13	

NOTES:

1. For First Article Group B testing, measure the impedance of all 16 CBS inductor elements, at 200 Hz, before installing in equipments. For production Group B testing only, the resistance is measured per paragraph 4.8.7.

2. For First Article Group B testing, the minimum audible alarm level is measured only on one unit. See 4.8.11.2.1. For production Group B testing, the level is measured per paragraph 4.8.11.2.2.

3. For First Article Group B testing, the Surge Protection test is performed only on the nuclear survivability unit #6. The Surge Protection test is not performed on production Group B units.

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Table VII Group C Inspection

Inspection	Reqt Para	Insp Para
Subgroup 1		
Leakage (Immersion)	3.9.4	4.7.7
Temperature	3.9.1	4.7.1
Altitude	3.9.3	4.7.2
Vibration	3.9.9	4.7.4.1
Leakage (Immersion) repeat	3.9.4	4.7.7
Humidity	3.9.2	4.7.3
Subgroup 2		
Shock, bench handling	3.9.6	4.7.5
Bounce, loose cargo	3.9.7	4.7.4.2
Shock, drop	3.9.5	4.7.6
EMI	3.10	4.10
Subgroup 3		
Dust (fine sand)*	3.9.10	4.7.8
Salt Fog*	3.9.8	4.7.9
Fungus*	3.9.11	4.7.10

*The equipment shall be thoroughly washed, cleaned, dried and refurbished after this inspection before proceeding with subsequent inspection.

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4.5.3.1.2 Subgroup 2. For this subgroup, two units shall be selected at random from the first production lot. For subsequent Group C inspection, two units shall be selected from three months production or 300 units, whichever is smaller. One unit shall be subjected to EMI testing and one unit shall be subjected to loose cargo bounce, bench handling shock and shock drop testing.

4.5.3.1.3 Subgroup 3. For this subgroup, two units shall be selected at random from the first production lot. For subsequent Group C inspection, two units shall be selected from six months production or 600 units, whichever is smaller. One unit shall be subjected to Dust and Salt Fog Testing; and one unit shall be subjected to Fungus Testing.

4.5.3.2 Order of Inspection within Group C. Group C inspection shall be performed in the order specified in Para 4.5.3.1.

4.5.3.3 Group C Failures. Actions required relative to Group C failures shall be as specified in the contract or purchase order.

4.5.4 Reinspection of Conforming Group B and Group C Sample Units. Unless otherwise specified sample units which have been subjected to and passed both Group B and Group C inspection may be accepted on contract provided all damage is repaired and the sample units are resubjected to and pass Group A inspection.

4.5.5 Group D Inspection. This inspection shall consist of the tests specified in Table VIII, and shall be performed on units from lots which have been subjected to and met Group A and Group B inspection.

4.5.5.1 Sampling for Group D Inspection. For reliability purposes ten samples shall be selected at random once each eight months production or 800 units, whichever is greater.

Table VIII Group D Inspection

Inspection	Reqt Para	Insp Para
Reliability	3.11	4.11

4.5.5.2 Group D Failures. Actions required relative to Group D failures shall be as specified in the contract.

4.5.5.3 Reinspection of Conforming Group D Sample Units. Unless otherwise specified, sample units which have been subjected to and passed Group D inspection may be accepted on the contract or purchase order provided all visible and known damage is repaired and the sample units are resubjected to and pass Group A inspection.

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4.6 Preconditioning (See 3.7)

4.6.1 Bounce Preconditioning. Each unit, in its non-operating condition, shall be placed in its normal operating position on the table of the package tester type 1000SC, as made by the L.A.B. Corporation, Skaneateles, NY, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32 inch thick strip of material under one corner or edge of the unit to a distance of 3 inches as the unit bounces. The unit shall be subjected to this preconditioning for 1 minute. Each unit of each lot shall be subjected to this preconditioning. After bounce preconditioning, the unit shall not be repaired, aligned, cleaned, or otherwise changed prior to subjection to acceptance inspection (See 3.7.1).

4.6.2 Electrical Preconditioning (burn-in). Each unit TA-978( )/G shall be subjected to the test conditions cited in 4.11.3 and shall meet the requirements of 3.7.2. At 24 hours in test the operational performance test of paragraph 4.15 shall be performed. At completion of test (at 48 hours) the tests of paragraphs 4.15 and 4.16 shall be performed. (See 3.7.2).

4.7 Environmental Tests. The following tests shall be performed on the equipment. (See 3.9).

4.7.1 Temperature Tests. (See 3.9.1).

4.7.1.1 Low Temperature. The equipment shall be subjected to the test of Method 502.1, Procedure I, of MIL-STD-810. The storage temperature (step 2) shall be -70^{0} F (-56.7^{0}C) and shall be maintained for a period of not less than 2 hours following stabilization of the test item. Low operating temperature (step 4) shall be -30°F (-34.4^{o}C). (See 3.9.1.1).

4.7.1.2 High Temperature. The equipment shall be subjected to the test of Method 501.1, Procedure II, of MIL-STD-810. For step 4 the chamber temperature shall be $+160^{o}$F ($+71^{o}$C). For step 7 the chamber temperature shall be adjusted to $+150^{o}$F ($+65^{o}$C) for test item operation (this includes effects of solar radiation). (See 3.9.1.2).

4.7.2 Altitude Test (See 3.9.3). The equipment shall be tested in accordance with Method 500.1, Procedure I of MIL-STD-810. For step 2 the chamber altitude shall be 15,000 feet above sea level. In addition, the equipment shall withstand air transportation altitude up to 40,000 feet above sea level while nonoperating.

4.7.3 Humidity Test (See 3.9.2). The equipment shall be subjected to Test Method 507.1, Procedure II of MIL-STD-810. In step 6, measurements shall be taken during the last five (5) hours of the last cycle.

4.7.4 Vibration, Bounce and Shock Tests. (See 3.9.9, 3.9.7, 3.9.5, 3.9.6).

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4.7.4.1 Vibration. (See 3.9.9). The equipment shall be subjected to the test of Method 514.2, Procedure VIII, Figure 5.4.2-6 Curve V of MIL-STD-810. The time schedules shall be in accordance with Table 5.4.2-VI of MIL-STD-810. The equipment shall be electrically operated and mechanically inspected before and after the test and the two results compared.

4.7.4.2 Bounce, Loose Cargo (See 3.9.7). The equipment in the field portable (loose cargo) configuration shall be subjected to Test Method 514.2, Procedure XI, Part 2 of MIL-STD-810.

4.7.5 Shock Test - Bench Handling. (See 3.9.6). The equipment shall be subjected to Test Method 516.2, Procedure V of MIL-STD-810.

4.7.6 Shock Test Drop. (See 3.9.5). The equipment shall be subjected to Test Method 516.2, Procedure II of MIL-STD-810.

4.7.7 Immersion Test. (See 3.9.4). The equipment shall be subjected to Test Method 512.1, Procedure I of MIL-STD-810.

4.7.8 Dust Test. (See 3.9.10). The equipment shall be subjected to Test Method 510.1, Procedure I of MIL-STD-810.

4.7.9 Salt Fog Test. (See 3.9.10). The equipment shall be subjected to Test Method 509.1, Procedure I of MIL-STD-810. Operation of the equipment shall not be required and failure criteria shall be limited to corrosion of finishes and metals only. Such corrosion shall be defined as any visible degradation of the equipment surfaces that can be attributed to flaky, pitted, blistered or otherwise loosened finish on metal surface.

4.7.10 Fungus Test. (See 3.9.11). The equipment shall be subjected to Test Method 508.1, Procedure I, of MIL-STD-810. The equipment shall be opened and the parts and materials of the interior circuit assemblies shall be sprayed with spore suspension. The interior and exterior of the case shall also be sprayed. Operation is not required. Inability of the equipment to meet the requirements of 3.9.11 shall constitute failure of the test.

4.8 Electrical Tests (See 3.8).

4.8.1 Standard Conditions. Unless otherwise specified, all tests shall be performed with the TA-978( )/G powered from a normal 28 Vdc power supply or battery BA-3590 or BA-5590, with the Signal Adapter (see figure 5) connected to the TA-978( )/G (see figure 6) and at room ambient temperature, pressure and humidity. Unless otherwise specified, the TA-978( )/G hookswitch is in the "off-hook" position, the TA-978( )/G MODE switch is in the CBS position, and the TA-978( )/G TEST switch is in the LIGHTS ON position, and the TA-978( )/G power switch is in the ON position.

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4.8.2 Test Equipment. Unless otherwise noted, contractor shall be responsible for providing all standard and special test equipment required to perform the quality assurance tests of paragraph 4.0. The following equipments shall be provided by the Government to the contractor as Government loaned equipment (GLE) in sufficient quantities and in a timely fashion compatible with the contractual schedule for the performance of the tests.

a. Battery Case Z-AIJ (ON241787)
b. Vehicular Power Adapter HYP-57 (ON 241698)
c. Battery BA-5590/U
d. Handsets H-250/U and H-189/GR

The following item shall be made available by the Government to the contractor:

Use of an anechoic chamber at a Government facility to perform test 4.8 .11..2.1.

4.8.3 Plain Text Impedances. (See 3.8.3)

4.8.3.1 Wire Line Interface Impedance. (See 3.8.3.1)

4.8.3.1.1 Receive Condition. Set Signal Adapter MODE switch to PT. Set Signal Adapter PTT switch to OFF. Set Signal Adapter WIRE LINE switch to OPEN(EXT). Connect Impedance Bridge to wire line terminals of TA-978( )/G, E1 and E2 balanced to ground. Apply 1 volt RMS at 1kHz frequency and measure impedance.

4.8.3.1.2 Transmit Condition. Set Signal Adapter MODE switch to PT. Set Signal Adapter PTT switch to ON. Set Signal Adapter WIRE LINE switch to OPEN (EXT). Repeat measurement of 4.8.3.1.1.

4.8.3.2 Secure Voice Unit Interface Impedance. (See 3.8.3.2) Set Signal Adapter MODE switch to PT. Set Signal Adapter PTT switch to ON. Set Signal Adapter WIRE LINE switch to 604 ohms. Connect Impedance Bridge to Signal Adapter J1. Measure impedance with a 10 mV rms signal at 1 kHz frecuency.

4.8.4 Cipher Text Impedances.(See 3.8.4)

4.8.4.1 Wire Line Interface Impedance. (See 3.8.4.1)

4.8.4.1.1 Receive Condition. Set Signal Adapter MODE switch to CT. Set Signal Adapter PTT switch to OFF. Set Signal Adapter WIRE LINE switch to OPEN (EXT). Connect Impedance Bridge to wire line terminals of TA-978( )/G. Measure impedance with 1V rms signal at 16 kHz frequency.

4.8.4.1.2 Transmit Condition. Set Signal Adapter MODE switch to CT. Set Signal Adapter PTT switch to ON. Set Signal Adapter WIRE LINE switch to OPEN (EXT). Repeat measurement of 4.8.4.1.1.

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4.8.4.2 Secure Voice Interface Impedance. (See 3.8.4.2). Set Signal Adapter MODE switch to CT. Set Signal Adapter PTT switch to ON. Set Signal Adapter WIRE LINE switch to 150 ohms. Connect Impedance Bridge to Signal Adapter J2. Measure impedance with 1V rms signal at 16 kHz.

4.8.5 Ringer Output. (See 3.8.5). Set Signal Adapter MODE switch to PT set Signal Adapter PTT switch to OFF. Set Signal Adapter WIRE LINE switch to 1500 ohms. Set TA-978( )/G MODE switch to RD. Connect to Signal Adapter J5: RMS Voltmeter, Oscilloscope, Frequency Counter (through 10:1 attenuator).

4.8.5.1 Depress TA-978( )/G RING switch and observe 20 Hz squarewave on oscilloscope and measure rms voltage level ($52 < Vrms < 108$) and frequency (20 $\pm$4 Hz).

4.8.5.2 Set Signal Adapter wire line switch to EXT-R. Connect resistor 5,000 ohm/2%/3W to EXT-R terminal. Repeat 4.8.5.1 and measure rms voltage ($72 < Vrms < 108$) and frequency (20 $\pm$4 Hz)

4.8.5.3 Connect resistor 100,000 ohm/5%1W to EXT-R terminal and repeat 4.8.5.1. Measure rms voltage ($72 < Vrms < 108$) and frequency (20 $\pm$ 4 Hz). At completion of test, set TA-978( )/G MODE switch to CBS. Depress TA-978( )/G RING switch and observe and verify that ring signal is no longer present.

4.8.6 Ring Detector (See 3.8.6). Set Signal Adapter MODE switch to PT. Set Signal Adapter PTT switch to OFF. Depress and hold TA-978( )/G hookswitch to "on-hook" and alarm volume to high. Apply 25 volt RMS, 20 Hz to WIRE LINE terminals and verify that visual (amber) CALL indicator is lit and that audible alarm sounds. Verify proper operation of volume control from low (barely audible) to high. (See paragraph 4.8.11.2).

4.8.7 Common Battery Supervision. (See 3.8.7). Set Signal Adapter MODE switch to PT. Set Signal Adapter PTT switch to OFF. Set Signal Adapter WIRE LINE switch to OPEN (EXT). Verify that TA-978( )/G hookswitch is "off-hook". Connect resistance meter to Signal Adapter J5 and measure dc resistance of CBS element ($<$450 ohms). Depress hookswitch and again measure dc resistance ($>$100,000 ohms).

4.8.8 DTMF Signaling. (See 3.8.2)

4.8.8.1 Wire Line Signal Level and Frequencies. (See 3.8.2.1 and 3.8.2.2). Set Signal Adapter MODE switch to PT. Set Signal Adapter PTT switch to OFF. Set Signal Adapter WIRE LINE switch to 604 ohms. Connect to Signal Adapter J5. RMS Voltmeter, Oscilloscope, Frequency counter. Depress TA-978( )/G keysender keys as shown in table IX and verify presence of dual and single tones with oscilloscope, and measure dual tone signal levels and single tone frequencies as indicated.

4.8.8.2 Sidetone Signal Level. (See 3.8.2.3). Keep conditions as in 4.8.8.1. Connect RMS Voltmeter to Signal Adapter J3. Depress key P and measure signal level (70 $\pm$16 mVrms, -21 $\pm$ 2dBm).

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Table IX DTMF Signaling

	Level ($0.48\ ^{+0.12}_{-0.1}$ V_{rms}) (-4 ±2 dBm)	Frequency (±1.3%)
1	_____	
1 & 2		_____ (697 ±9)
1 & 4		_____ 1209 ±17)
2	_____	
2 & 5		_____ (1336 ±17)
3	_____	
3 & 6		_____ (1477 ±19)
4	_____	
4 & 5		_____ (770 ±10)
5	_____	
6	_____	
7	_____	
7 & 8		_____ (852 ±11)
8	_____	
9	_____	
0	_____	
0 & R		_____ (941 ±12)
F0	_____	
F0 + F		_____ (1633 ±21)
F	_____	
I	_____	
P	_____	
R	_____	
C	_____	

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4.8.9 Plain Text Signal Characteristics. (See 3.8.8).

4.8.9.1 TA-978( )/G to Wire Line. (See 3.8.8.1).

4.8.9.1.1 Signal Level, Frequency Response and Distortion. (See 3.8.8.1.1). Set Signal Adapter MODE switch to PT. Set Signal Adapter PTT switch to On. Set Signal Adapter WIRE LINE switch to 604 ohms. Connect to Signal Adapter J1. Audio Signal Generator, RMS Voltmeter, Distortion Analyzer. Connect to Signal Adapter J5. RMS Voltmeter, oscilloscope (for observation only), Distortion Analyzer. Apply a 1 millivolt rms sinewave signal of 1 kHz with maximum 1% distortion to Signal Adapter J1 and measure wire line signal level (reference level A) and distortion, see Table X.

Repeat measurements at three other frequencies as listed in Table X.

Table X Plain Text Wire Line Signals

Frequency	Level	Distortion
1 kHz (Ref.)	_____=A(-4 ±3 dBm) (.34 to .68 Vrms)	_____5%
300 Hz	_____(A ±2 dB)	_____5%
2 kHz	_____(A ±2 dB	_____5%
3.5 kHz	_____(A ±2 dB)	_____5%

4.8.9.1.2 Sidetone Signal Level. (See 3.8.8.1.2). Keep conditions as in 4.8.9.1.1 with 1 kHz input signal. Connect RMS Voltmeter to Signal Adapter J3. Measure signal level (-12 ± 2 dB below signal level A).

4.8.9.2 TA-978( )/G to Secure Voice Device. (See 3.8.8.2).

4.8.9.2.1 Signal Level Frequency Responses and Distortion. (See 3.8.8.1.1). Set Signal Adapter MODE switch to PT. Set Signal Adapter PTT switch to OFF. Set Signal Adapter WIRE LINE switch to OPEN (EXT). Connect to Signal Adapter J5: Audio Signal Generator, RMS Voltmeter, Distortion Analyzer. Connect to Signal Adapter J3: oscilloscope (for observation only), RMS Voltmeter, Distortion Analyzer. Apply a +1 dBm (0.87 Vrms) sinewave signal of 1 kHz, with maximum 1% distortion, to Signal Adapter J5 and measure signal level (reference level B) and distortion at Signal Adapter J3, see Table XI, Apply an input level of -20 dBm (77.5 millivolts rms) and measure signal level (reference level A) and distortion see Table XI. Repeat measurements at both levels at three other frequencies as listed in Table XI

4.8.10 Cipher Text Signal Characteristics. (See 3.8.9).

4.8.10.1 TA-978( )/G to Wire Line. (See 3.8.9.1).

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4.8.10.1.1 Signal Level and Risetime. (See 3.8.9.1.1). Set Signal Adapter MODE switch to CT. Set Signal Adapter PTT switch to ON. Set Signal Adapter WIRE LINE switch to 150 ohms. Connect to Signal Adapter J2: Squarewave Generator, oscilloscope. Connect to Signal Adapter J5: oscilloscope. Apply 12V peak-to-peak squarewave signal of 16 kHz to Signal Adapter J2. Measure at Signal Adapter J5: Signal level (3.0 ± 0.5V peak-to-peak) and Risetime (maximum 10 microseconds).

NOTE: The preemphasis of the cipher signal driving circuit (which compensates for the capacitive load of a wire line) causes the signal to be "overpeaked" when fed into a 150-ohm resistive load. Measure the signal by averaging as shown in figure 2.

Table XI. Plain Text Signal to Secure Voice Unit

FREQUENCY	INPUT	OUTPUT LEVEL	DISTORTION
1 kHz (Ref)	-20 dBm	_____=A (-20 ±1.5 dBm)	_____(5%)
1 kHz (Ref)	+ 1 dBm	_____=B (+1 ±1.5 dBm)	_____(5%)
300 Hz	-20 dBm	_____(A ±2 dB)	_____(5%)
300 Hz	+ 1 dBm	_____(B ±2 dB)	_____(5%)
2 kHz	-20 dBm	_____(A ±2 dB)	_____(5%)
2 kHz	+ 1 dBm	_____(B ±2 dB)	_____(5%)
3.5 kHz	-20 dBm	_____(A ±2 dB)	_____(5%)
3.5 kHz	+ 1 dB m	_____(B ±2 dB)	_____(5%)

4.8.10.2 TA-978( )/G to Secure Voice Device (See 3.8.9.2)

4.8.10.2.1 Low Signal Level. (See 3.8.9.2.1). Set Signal Adapter MODE switch to CT. Set Signal Adapter PTT switch to OFF. Set Signal Adapter WIRE LINE switch to OPEN (EXT). Connect to Signal Adapter J5: Squarewave Generator, oscilloscope. Connect to Signal Adapter J4: oscilloscope. Apply a 15 millivolt peak-to-peak squarewave signal of 16 kHz to Signal Adapter J5. Measure at Signal Adapter J4, signal level in volts peak-to-peak.

NOTE: Due to capacitive coupling of signal in the TA-978( )/G, the signal waveform is as shown in figure 3. Measure by averaging as shown.

4.8.10.2.2 High Signal Level. (See 3.8.9.2.2). Keep conditions as in 4.8.10.2.1. Apply a 3.5V peak-to-peak squarewave signal of 16 kHz to Signal Adapter J5.

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NOTE: Due to limiting of signal in the TA-978( )/G, the high level output signal is a squarewave.

4.8.10.2.3 Distortion. (See 3.8.9.2.3). Set Signal Adapter as in 4.8.10.2.1. Connect to Signal Adapter J5: Audio Signal Generator, RMS Voltmeter, oscilloscope (for observation only). Connect to Signal Adapter J4: RMS Voltmeter, Distortion Analyzer. Apply a 15 millivolt rms sinewave signal of 10 kHz to Signal Adapter J5. Measure the distortion at Signal Adapter J4, see Table XII. Repeat measurements at other frequencies as listed in Table XII.

Table XII. Cipher Signal Distortion

Frequency (kHz)	Distortion
10	_____(5%)
20	_____(5%)
50	_____(5%)
76	_____(5%)
100	_____(5%)

4.8.11 Audible and Visual Alarms (See 3.8.1).

4.8.11.1 Visual. (See 3.8.1.2) Set Signal Adapter MODE switch to PT. Set Signal Adapter PTT switch to OFF. Set Signal Adapter WIRE LINE switch to OPEN (EXT). Set TA-978( )/G ALARM Volume to High. See paragraph 4.8.6 for test of CALL indicator and audible alarm with external ring signal. Push TA-978( )/G TEST switch to TEST position. Verify that visual (amber) CALL indicator is lit.

4.8.11.2 Audible. (See 3.8.1.1). Keep the same conditions as in 4.8.11.1. Push TA-978( )/G TEST match to TEST position. Verify that audible alarm sounds and verify that ALARM volume controls the sound level from low (barely audible) to high.

4.8.11.2.1 Alarm Volume Measured in an Anechoic Room. The following test is performed on one First Article unit TA-978( )/G as part of the First Article Group B tests. Install a TA-978( )/G with battery attached in an anechoic room. Install Sound Measuring device at three feet distance from TA-978( )/G front panel. Operate TA-978( )/G and push TEST switch to TEST position. Set ALARM volume to LOW and measure sound level (32 $\pm$ 5 dBa). Set ALARM volume to HIGH and measure sound level ($>$88 dBa).

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4.8.11.2.2 Alarm Volume Measured for Production Equipments Group B Tests.

a. High level. Set ALARM volume to HIGH and hold microphone of Sound Measuring device directly against alarm cavity grille on front panel of TA-978( )/G. Push TEST switch to TEST position. Measure sound level with Sound Measuring device (130 ± 5 dBa).

b. Low level. Loosen and remove front panel assembly 1A3 of TA-978( )/G and reconnect to unit with extender board SM-D-944457. Connect RMS Voltmeter to terminals E17 and E18 of 1A3A1 printed wiring assembly. Set ALARM volume to LOW and push TEST switch to TEST position. Measure RMS voltage at terminals E17 and E18 (40 ± 10 millivolts rms).

c. Alarm tone frequency. Keep condition of b. above. Connect Frequency Counter between test jack TP1 and ground of 1A3A1 printed wiring assembly. Push Test switch to TEST position. Measure tone frequency of alarm (1850 ± 100 Hz).

4.8.11.3 Cipher Text Indicator. (see 3.8.1.2.2). Set Signal Adapter MODE switch to CT. Set Signal Adapter PTT switch to OFF. Set Signal Adapter WIRE LINE switch to OPEN (EXT). Verify that visual (green) CIPHER indicator is lit. Set Signal Adapter PTT switch to ON. Verify that visual (green) CIPHER indicator is lit. Set Signal Adapter MODE switch to PT. Verify that visual (green) CIPHER indicator is off. Set TA-978( )/G TEST switch to TEST. Verify that both amber and green indicator lights are lit.

4.8.12 Power Requirements. (See 3.8.10.).

4.8.12.1 Battery Voltage. (See 3.8.10.1). Operate the TA-978( )/G from 20 Vdc and verify proper operation with the Operational Performance Test (See paragraphs 3.8.11 and 4.15, Table XIII), measure supply currents as indicated in Table XIII. Operate the TA-978( )/G from 32 Vdc and verify proper operation with the Operational Performance Test per Table XIII: measure supply currents as indicated in Table XIII.

4.8.12.2 Battery Power. (See 3.8.10.2).

4.8.12.2.1 Maximum Power Average. (See 3.8.10.2.1). Calculate the maximum average power, with the use of the supply currents measured at 32 Vdc in paragraph 4.8.12.1, in accordance with the formula of Note 7 of Table XIII (maximum 3.0 watts).

4.8.12.2.2 Standby Power. (See 3.8.10.2.2). Press hookswitch of TA-978()/G down and keep in "on-hook" position. Adjust power source to 28 Vdc and measure supply current (0 mA). Adjust power source to 32 Vdc and measure supply current (maximum 15 mA).

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4.8.12.3 Vehicular Power. (See 3.8.10.3). Operate the TA-978( )/G, with the vehicular Power Adapter HYP-57 attached, from a 22 Vdc source, and verify proper operation with the Operational Performance Test (See paragraph 3.8.11 and 4.15, Table XIII). Repeat the Operational Performance Test with a supply source voltage of 32 Vdc.

4.8.12.4 Transient Voltage Protection. (See 3.8.10.4). Operate the TA-978( )/G with the HYP-57 attached from a 28 Vdc source. Perform test of MIL-STD-1275(AT) paragraphs 6.2.2 and 6.2.3. Operate the TA-978( )/G, with the HYP-57 attached from a 40 Vdc source. Perform test of MIL-STD-1275(AT), paragraphs 6.2.4.c. During these tests perform Operational Test of Table XIII, steps 3, 5, 6, 7, 8 and 9, and observe in each step the output signal of the TA-978( )/G for evidence of malfunction.

4.8.12.5 NESTOR Power. (See 3.8.10.5). Switch Signal Adapter NESTOR power switch to ON and measure the dc voltages at Signal Adapter test jacks TP1 (9 Vdc) and TP2 (5 Vdc).

4.8.12.6 Reverse Polarity Protection. (See 3.8.10.6). Operate the TA-978( )/G with the HYP-57 attached from a 28 Vdc source voltage. Reverse the polarity of the supply and keep connected for five minutes. Return polarity to normal and verify proper operation with the Operational Performance Test of Table XIII.

4.8.13 Surge Protection. (See 3.8.10.7). The following test is performed on one First Article unit TA-978( )/G as part of the First Article Group B tests. Apply a current surge as shown in figure 4 to each wire line terminal of the TA-978( )/G and to both terminals connected together, with the TA-978( )/G in operating condition. After test, verify proper operation with the Operational Performance Test of Table XIII. This Surge Protection test is not performed during production equipment Group B tests.

4.9 Mechanical Characteristics. (See 3.6).

4.9.1 Inspection for Dimensional Interchangeability. The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirement of 3.6. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. External and interdimensions of cavities, when such dimensions of cases, covers, and insertable assemblies affect mating of parts.

b. Dimensions of cavities, when such dimensions affect insertion of items.

c. Location of fasteners on separable parts or assemblies which must mate, such as cases, covers, and mountings.

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d. Location of connectors, locking pins, fasteners, and mountings which receive mating parts of plug-in assemblies and major units; and location of the mating parts on the plug-in assembly or major unit.

e. Size and form of special threads.

f. Dimensional relationship between enclosure and printed wiring assembly 1A1 when attached to enclosure.

g. Dimensional relationship between front panel and printed wiring assembly 1A3 when attached to front panel.

h. The interface dimensions at rear of enclosure with battery cases and battery.

4.9.2 Air Vent. (See 3.9.3). Pressure equalization through the air vent is verified during the altitude test of 4.7.2 by observing that, during the pressure variations in the chamber, any bulge or depression of the rubber cover of the keysender returns to normal within five minutes.

4.10 Electromagnetic Interference. (See 3.10).

4.10.1 Electromagnetic Compatibility Test. (See 3.10.1). The TA-978( )/G shall be tested for compliance with the requirements of 3.10.1. Test setups and test procedures shall conform to the measurement techniques of MIL-STD-462, Notice 3. Signal and data input and output terminals shall be terminated with simulated inputs and loads. Simulated loads shall have characteristic impedance similar to actual loads. All cables used during test shall be of the same type and length to be supplied by the contractor or used in the actual installation. All electromagnetic compatibility tests shall be performed with the vehicular power supply installed.

4.10.2 Bonding Impedance Test. (See 3.10.2). Prior to the performance of the EMI/EMC testing of 4.10, bonding impedances shall be measured for compliance to the requirements of 3.10.2. All measurement readings shall be recorded and included in the EMI/EMC test report. Measurements shall be performed from the equipment case to the shell of case-mounted connectors to all other externally mounted connectors. Additionally, measurements shall be performed between all mating surfaces bonded by EMI gasketing.

4.11 Reliability. (See 3.11).

4.11.1 First Article Reliability Test Length/Accept Reject Criteria. The reliability of the equipment shall be demonstrated by selecting ten (10) First Article units TA-978( )/G that have passed Group A and Group B inspection and subjecting them to a total of 12,065 operating hours. The accept/reject criteria is:

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Accept on 2 or less failures.

Reject on 3 or more failures.

Test conditions shall be as specified in paragraph 4.11.3. No single equipment "on" time shall be less than one-half the average operating time of all equipments on test.

4.11.1.1 Pattern Failures. Occurrence of a pattern failure (para 3.1.4.5, MIL-STD-781C) shall be cause for rejecting the lot.

4.11.2 Quality Conformance (Group D) Reliability Test Length/Reject Criteria. The reliability of production equipment shall be demonstrated by subjecting them to 4,000 operating hours. The accept/reject criteria are:

Accept on 0 failures.

Reject on 1 or more failures.

A quantity of ten samples shall be selected from lots that have passed Group A and Group B inspection. Test conditions shall be as specified in paragraph 4.11.3. No single equipment "on" time shall be less than one half the average operating time of all equipments on test.

4.11.3 Test Conditions

4.11.3.1 Duty Cycle. The testing shall be conducted on a 24 hour, 7 day per week basis.

4.11.3.2 Temperature. The equipment shall be operated in an ambient temperature of 50°C + 5°C - 0°C (122°F to 131°F).

4.11.3.3 Vibration. The equipment shall be subjected to a vibration of 2.2g ± 10% at a nonresonant frequency between 20 and 60 Hz. The vibration shall be run for 10 minutes during each operating hour.

4.11.3.4 Input Power. The equipment shall be operated during the test from a nominal + 28 Vdc source.

4.11.3.5 Input Power On-Off Cycling. The input power shall be cycled between 2 hours 45 minutes on and 15 minutes off during the length of the test.

4.11.3.6 Ring Signal Operating. The equipments shall be operated in the ringdown mode and the ring switch of each unit shall be engaged by automatic means for six seconds out of each minute. The ring signal shall be applied to a resistive load of 1500 ohms ± 5%, 5W. A permanent recording of the presence of the ring signal shall be made for each unit during the entire test.

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4.11.3.7 Keypad Exercise. A keypad shall be exercised every 3 hours of "on-time" per day by mechanical means. On a daily basis, the keypad exercisor shall move from one unit to another in serial number sequence so that all test samples are activated during the duration of the demonstration. The keypad exercisor shall activate five (5) keys sequentially which shall be #1, #2, #4, R and P for a period of 15 minutes at a repetitive rate for each key of twice per minute. The activation time of each key shall be 5 seconds. (See Figure 7, Time Sequence for Keypad Exercisor).

4.11.3.8 Performance Parameters to Measure

4.11.3.8.1 Operational Performance Test. (See para 3.8.11 and 4.15) Performance once every 24 hours on each unit and results recorded.

4.11.3.8.2 Performance Test (See 4.16). Performed once every 24 hours on one unit, with a different unit selected each day, in sequence of serial numbers, with a recording of the test results.

4.11.3.9 Reliability Failures. The following requirements apply to both the reliability qualification tests and the production reliability sampling test.

4.11.3.9.1 Failure Actions, Categories and Confirmation. Failure actions categories, and verification shall be as specified in paragraphs 5.6 and 5.8 of MIL-STD-781.

4.11.3.10 Preventive Maintenance. No preventive maintenance shall be performed on the units during the reliability demonstration. However, operator controls shall be adjusted as required.

4.11.3.11 Test Reports and Failure Reporting and Analysis. Test reports, failure reporting and failure analysis shall be as specified in the contract.

4.12 Nuclear Survivability Tests. (See 3.12). Nuclear Tests (blast, thermal, EMP, Initial Nuclear Radiation) shall be performed on the equipment in accordance with MIL-N-49205(CR). The equipment shall meet the operational performance test of paragraph 3.8.11 following each test.

4.13 Visual and Mechanical Inspection. (See 3.6 and 3.17). The equipment shall be examined for the defects listed in MIL-STD-252. An inspection shall be performed to verify compliance with those portions of safety engineering which can be determined visually.

4.14 Quality Conformance Inspection and Packaging. Packaging shall be inspected in accordance with MIL-P-116 to determine conformance to the requirements of Section 5.

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4.15 Operational Performance Test. (See 3.8.11). All equipments to be operationally inspected shall be subject to the tests of Table XIII.

4.16 Performance Tests. (See 4.11.3.7.2). During the performance test the equipments are subjected to the tests of Table XIV.

Table XIII. Operational Performance Test

Test No.	Hook Switch Position	Test Para Reference	Supply Voltage I (Note 1)	Supply Voltage II (Note 2)	Supply Current at 32 Vdc (Notes 3,7)
1	On-hook	4.8.12.2.2		32	Measure
2	On-hook	4.8.6	28	32	Measure
3	Off-hook	4.8.5.1	28	32	Measure (A)
4	Off-hook	4.8.7	-	-	-
5	Off-hook	4.8.8.1 (note 4)	28	32	-
6	Off-hook	4.8.9.1.1 (Note 5)	28	32	Measure (B)
7	Off-Hook	4.8.9.2.1 (Note 6)	28	32	Measure (C)
8	Off-hook	4.8.10.1.1	28	32	Measure (D)
9	Off-hook	4.8.10.2.1	28	32	Measure
10	Off-hook	4.8.10.2.2	28	32	Measure (E)
11	Off-hook	4.8.11.2.2.a (Note 8)	28	-	-
12	Off-hook	4.8.12.5	28	32	-

NOTES:

1. Supply voltage used for all operational performance tests except when performed during reliability, temperature, and power tests.

2. Supply voltage used during operational performance test periods of the reliability test.

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3. Current measurements are made only during reliability, temperature, and power tests.

4. Frequencies of 697 Hz and 1209 Hz only. Signal level of key P only.

5. At 1 kHz frequency only, no distortion measurement.

6. At 1 kHz frequency only and +1 dBm input level, no distortion measurement.

7. Calculate maximum average power (see paragraph 3.8.10.2.1)as follows:

$$\left(\frac{\text{Sum of currents B+C+D+E}}{4} + \frac{\text{Current A}}{10}\right) \times 32 = \text{max. avg. power}$$

8. This test is not performed on units inside reliability and service condition test chambers.

4.17 Air-Seal Test. (See 3.13). The TA-978( )/G shall be opened and closed again in such a manner as to break and remake the seal. The air equalization vent shall be blocked by suitable means. Immediately thereafter, the equipment as field transported shall be subjected to a vacuum of 1 pound per square inch (1 pound per square inch less than the atmospheric pressure surrounding the equipment) applied to the interior of the equipment enclosure. The vacuum then shall be valved-off and the interior pressure measured during the ensuing period of 1 minute. During this 1 minute period the decrease in vacuum shall not exceed 0.01 pound. The guage used for measurement of the vacuum shall be of such accuracy that a difference of 0.01 pound can be determined readily.

4.18 Electrical Performance of Assemblies. (See 3.16). Assemblies (modules) including enclosure assembly, shall be checked to determine compliance with the applicable test procedure drawing referenced in the drawing covering construction of the modules.

4.19 Rough Handling Test (Packaging). When the rough handling test, in accordance with MIL-P-116 is required by the contract, the Operation Test of 4.15 shall be conducted to determine freedom from malfunction caused by rough handling.

4.20 Specular Gloss. When measured in accordance with Method 6101 of Federal Test Method Standard 141, the specular gloss shall meet the requirements of 3.14.

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Table XIV. Performance Test

Test No.	Test Para Reference	Supply Voltage
1	4.8.3.1	28 Vdc
2	4.8.3.2	28 Vdc
3	4.8.4.1	28 Vdc
4	4.8.4.2	28 Vdc
5	4.8.8.1	28 Vdc
6	4.8.8.2	28 Vdc
7	4.8.9.1.1	28 Vdc
8	4.8.9.1.2	28 Vdc
9	4.8.9.2.1	28 Vdc
10	4.8.10.2.3	28 Vdc

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5. Packaging.

5.1. Preservation. Preservation shall be as specified in MIL-STD-726. coded as follows:

5.1.1 Level A. JR-1-1-ϕϕ-00-NS-X-ED-ϕ-00-A.

5.1.2 Level B. JR-1-1-ϕϕ-00-NS-X-ED-ϕ-00-B.

5.2 Packaging and Marking. Packing and marking shall be in accordance with MIL-E-55585.

6. NOTES

6.1 Intended Use. The Interface, Telephone Signaling TA-978( )/G operates as an Interface between secure voice or plain voice signals from TSEC equipments and Army switchboards. The TA-978( )/G transmits plain text or cipher text into wire lines to a switchboard or another TA-978( )/G, and it receives plain text or cipher text from a wire line and feeds the signals into TSEC equipment. The TA-978( )/G also provides the telephone signaling to access the switchboard or another TA-978( )/G unit.

6.2 Ordering Data. Procurement documents should specify the following:

a. Title, Number, and date of the specification and any amendment thereto.

b. Level A, Level B or commercial preservation and packing (See Section 5).

c. When First Article samples are required (See 4.3).

d. When the rough handling test is required (See 4.19).

6.3 Environmental. Environmental pollution measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity recommended disposability methods.

6.4 Group C Inspection. Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action. (See 4.5.3.3).

6.5 Location of Operational Inspection. It is desirable that the Operational Inspection (4.15) be performed at a location that will minimize handling (which might cause damage to the equipment) after this inspection is completed. It is recommended that the entire lot (including all previously inspected sample units) be sampled and inspected immediately prior to packaging.

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6.6 Nomenclature. The parenthesis in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example; TA-978(W)/G. The contractor should apply for a nomenclature in accordance with the applicable clause in the contract.

6.7 Verification Inspection. Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

6.8 Definitions.

6.8.1 Branching. Branching is a connected arrangement of filaments (Hyphol) formed by shoots or secondary items growing from the main stem or filament (Hypha).

6.8.2 Growth Colonization. Growth colonization is a mass of individual plants, generally of one species, living together; or a group of hyphae which is formed from one spore or cell and may be one individual plant. Colonization which completely covers the surface of the nutrient material constitute abundant growth.

6.8.3 Microbial Growth. Microbial growth is the growth of a very minute organism. Such organisms, when present in large numbers, may provide a colonly visible to the naked eye.

6.8.4 Sporulation. Sporulation is the formation of minute unicellular reproductive or dormant bodies, called spores.

6.8.5 Tubular Germination. Tubular germination is partial growth by the production of hyphae, which are tubular shaped fungae filaments. Tubular germination constitutes restricted individual spore growth not proceeding to colonization.

6.8.6 Exfoliation. Exfoliation is corrosion along the grain boundaries of the metal resulting in the peeling or separating, or both, of successive layers of the metal. The appearance resembles loose book pages or onion skin peeling.

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6.9 Preproduction Pack. When specified in the Bid request or contract, one acceptable model of Interface, Telephone Signaling TA-978( )/G, will be packed in a manner that will pass the First Article inspection prescribed in specification MIL-P-116. Illustrations of the packaging procedure, together with bills of material, will be prepared and furnished in accordance with the provisions of Signal Corps Drawing SC-A-362100.

Custodian

Army - (CR)

Preparing Activity

Army - (CR)

Project 5810-A040

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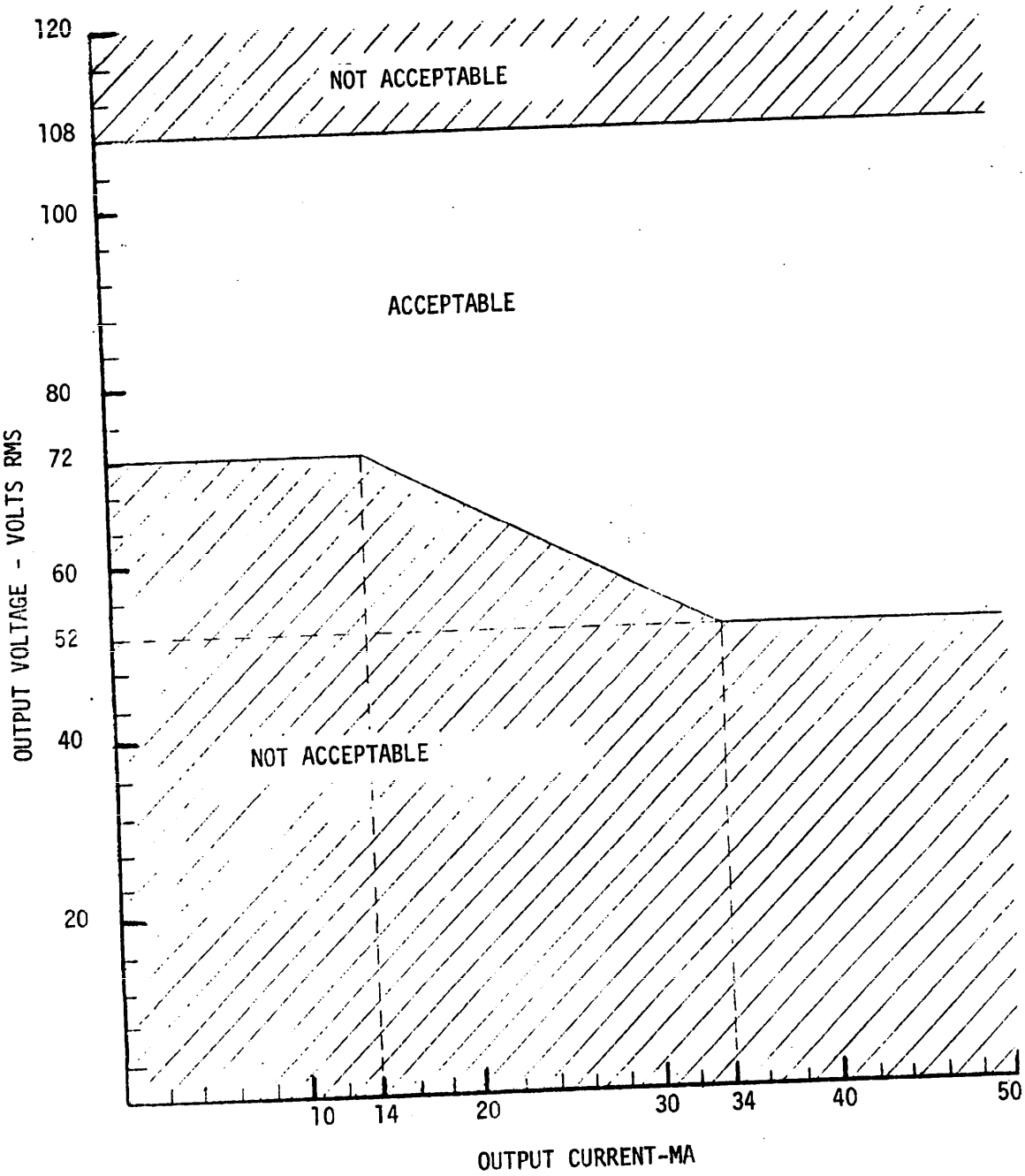
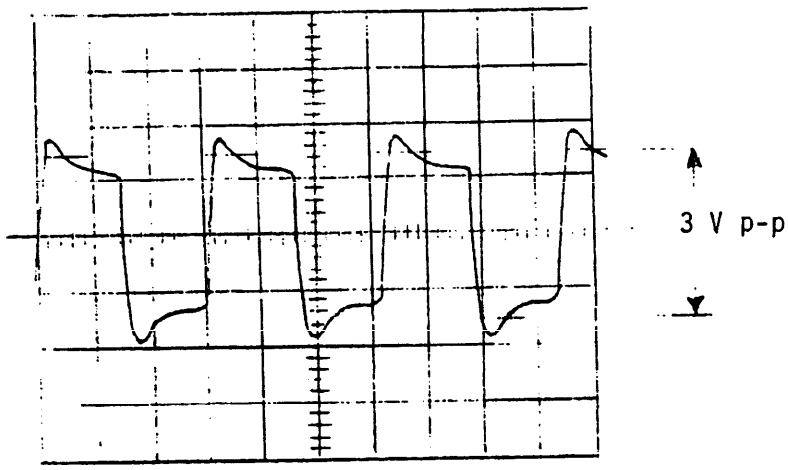


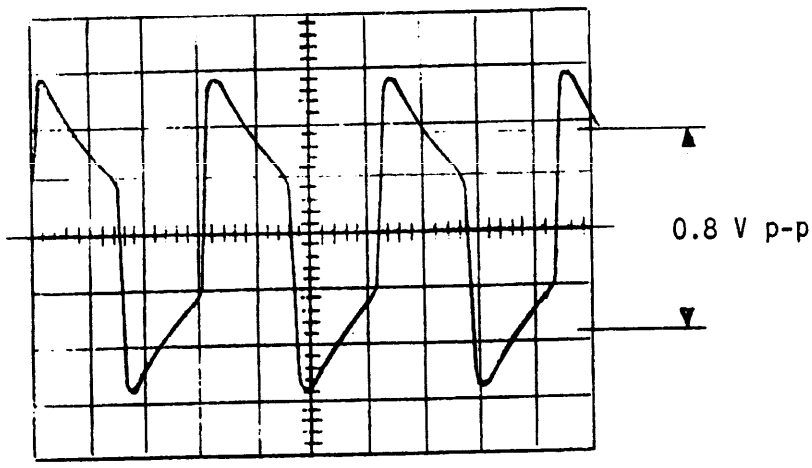
FIGURE 1 RING GENERATOR OUTPUT ENVELOPE

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Vertical 1 V/div

Figure 2



Vertical 0.2 V/div

Figure 3

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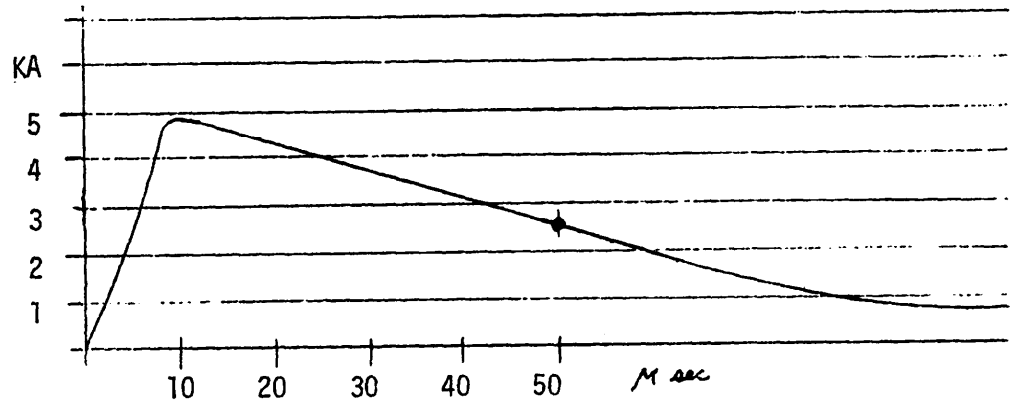


Figure 4. Surge Profile

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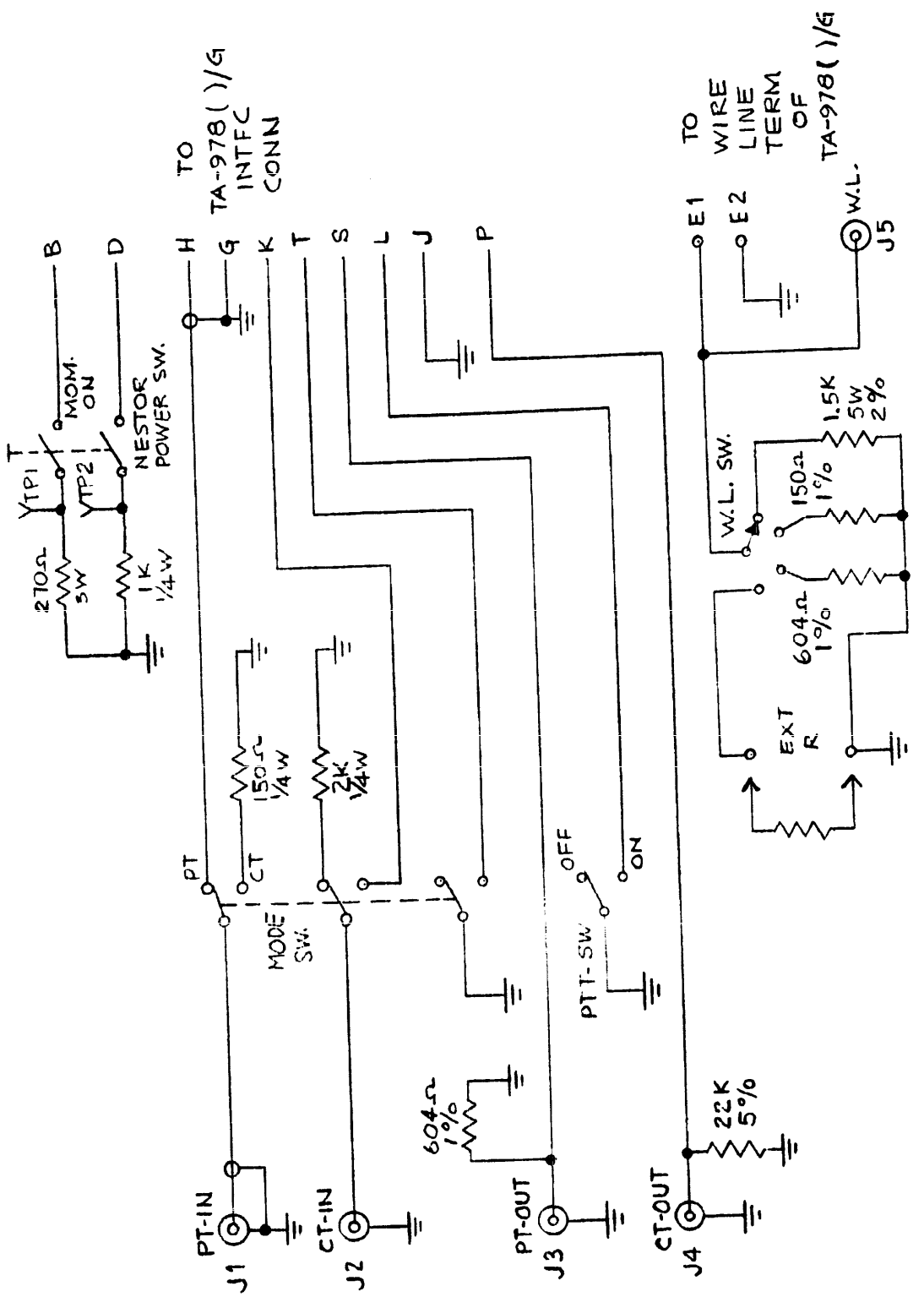


Figure 5. TA-978()/G Signal Adapter

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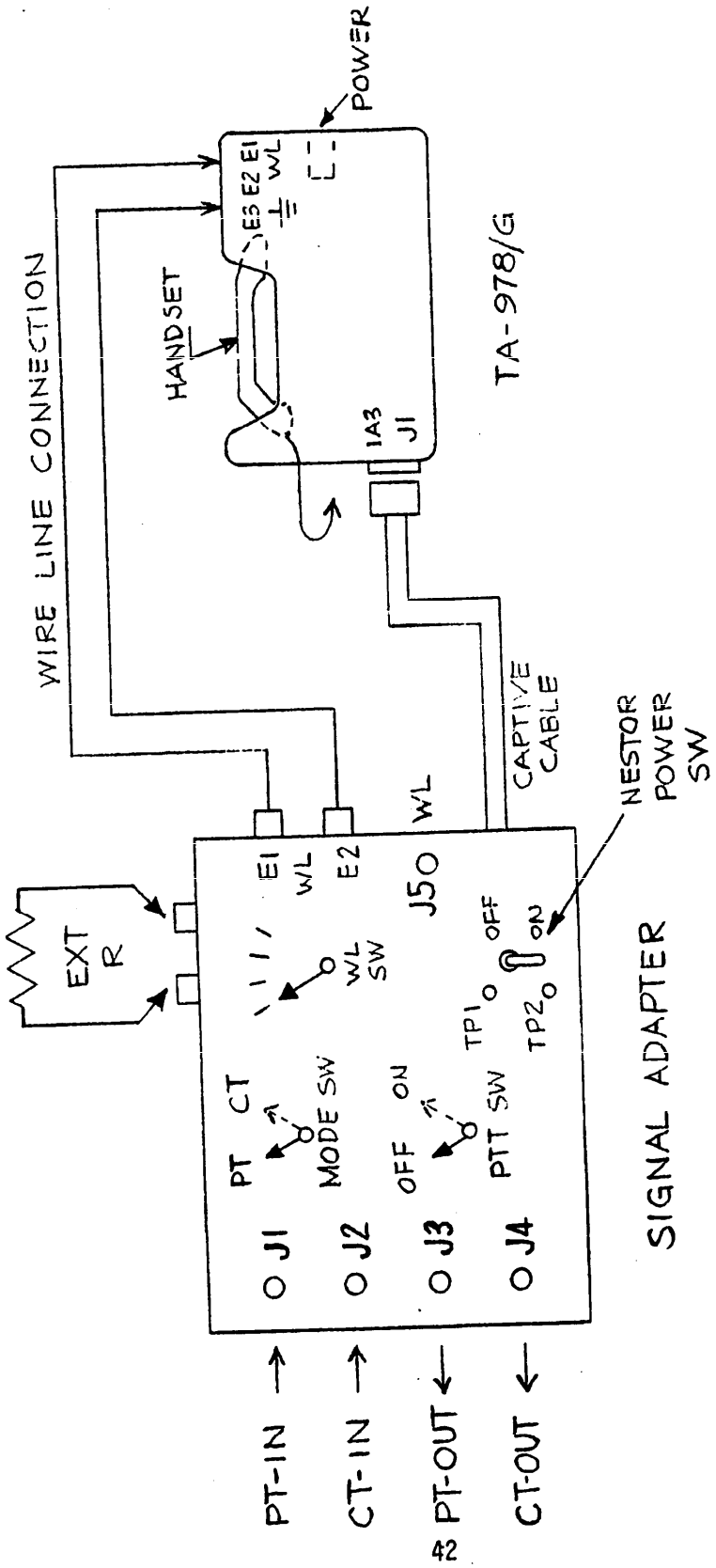
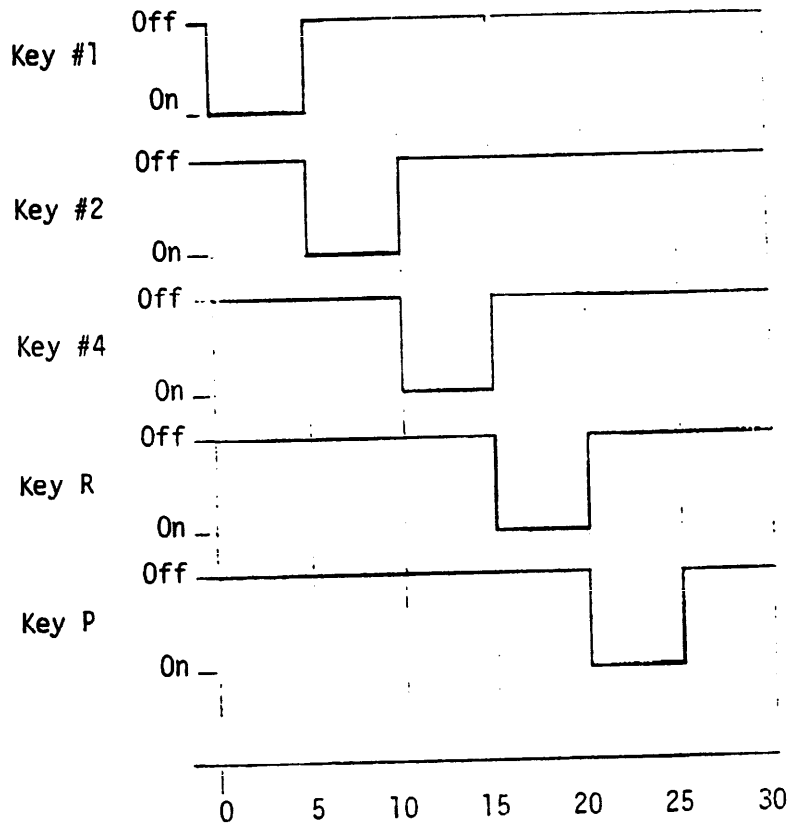


Figure 6. Signal Adapter - TA-978()/G Interconnection

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TIME SEQUENCE FOR KEYPAD EXERCISOR



(Activation Time in Seconds)

Figure 7

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
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MIL-I-49207 (CR)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

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☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

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   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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